

Chapter – 4

BLAST FURNACES

4.1 Introduction

BF is a counter current heat and mass exchanger, in which solid raw materials are charged from the top of the furnace and hot blast, is sent through the bottom via tuyeres. The heat is transferred from the gas to the burden and oxygen from the burden to the gas. Gas ascends up the furnace while burden and coke descend down through the furnace. The counter current nature of the reactions makes the overall process an extremely efficient one in reducing atmosphere. The real growth of blast furnace technology came with the production of high strength coke which enabled the construction of large size blast furnaces.

4.2 Raw materials and their quality

In India steel is being produced largely through the blast furnace. Iron ore, sinter and coke are the major raw materials for blast furnace smelting.

Raw materials:

The following **raw materials used** for the production of pig iron: -

- (i) Iron ore
- (ii) Limestone / L D Slag
- (iii) Dolomite
- (iv) Quartzite
- (v) Manganese ore
- (vi) Sinter
- (vii) Coke
- (viii) Pellets
- (ix) Scrap (Steel / Iron)
- (x) Coal Dust / Coal Tar

Iron ore: Iron bearing materials; provides iron to the hot metal. Iron ores is available in the form of oxides, sulphides, and carbonate, the oxide form known as hematite (red in colour) is mostly used in SAIL plants. It is the principal mineral in blast furnace for extraction of pig iron, generally rich in iron content varying from 62 % to 66 % associated often with naturally occurring fines (-10 MM) to the extent of 20 %. Although relatively free from impurities like phosphorous, sulphur and copper, they have high alumina and silica content as gangue. The high alumina content makes the slag highly viscous and creates problems for stable furnace operation.

Limestone / LD Slag: Acts as flux. Helps in reducing the melting point of gangue present in the iron bearing material and combines effectively with acidic impurities to form slag in iron making. LD slag is a substitute for limestone which is easily available in a steel plant. Its usage helps in waste utilization and thus reduces production cost.

Quartzite: It acts as an additive. Quartzite is a mineral of SiO_2 (silica) and under normal circumstances contains about 96 – 97 % of SiO_2 rest being impurities. Quartzite plays its role in counteracting the bad effects of high alumina in slag through maintaining optimum slag basicity.

Manganese ore: It acts as additive for the supply of Manganese in the hot metal. Manganese ore is available in the form of combined oxides of Mn and Fe and usual content of Mn is about 28 – 32 % for steel plant use, However Manganese ore available with SAIL is having high alkali contents so it should be used judiciously.

Coke: It acts as a *reducing agent* and fuel, supports the burden and helps in maintaining permeable bed. Coke (metallurgical) used in blast furnace both as fuel & reducing agent. The Indian coal is characterized by high ash (25 – 30 %) and still worse, a wide fluctuation in ash content, poor coke strength leading to excessive generation of fines, rapid fluctuation in moisture content etc. The problem of poor quality coke has been tackled by adding imported coal (75-95%) in the indigenous coal blend to get a coke ash of 13 – 16 %.

Sinter: It is iron bearing material. Fines that are generated in the plant/mines are effectively utilized by converting them to sinter. It provides the extra lime required for the iron ore and coke ash that is charged in the blast furnace. Sintering is the process of agglomeration of fines (steel plant waste and iron ore fines) by incipient fusion caused by heat available from the coke contained in the charge. The lumpy porous mass thus obtained is known as “sinter”.

Scrap (Steel / Iron): Scrap is generated in the process of product making in a steel plant which is gainfully utilized by back charging in the Blast Furnaces. It increases the furnaces productivity and reduces the production cost.

Pellets: It is also an iron bearing materials. The micro-fines which cannot be used for sinter making can be used for pellet manufacturing and the pellets formed will be charged in the BF.

Coal dust Injection: It acts as an auxiliary fuel, reduces coke consumption in the blast furnaces. The coal is injected through the tuyeres.

Different sources of raw materials

Sl. No.	Raw material	BSP	RSP	DSP	ISP	BSL
1.	Iron ore	Dalli Rajhara Raoghat Meghahatuburu Kiriburu	Barsua Kalta Meghahatuburu Kiriburu	Bolani GuaMeghaha tuburu	Gua Bolani Meghahatuburu	Kiriburu Meghahatuburu Bolani Barsua Gua Manoharpur
2.	Limestone	Nandini Kuteswar Jaisalmer Imported	Kuteswar Jaisalmer Imported	Kuteswar Jaisalmer Imported	Jaisalmer Imported	Nandini Kuteswar Jaisalmer Imported
3.	Dolomite	Hirri Imported	Baraduar Belha	Baraduar Imported	Belha Baraduar Imported	Birmitrapur Belha Imported

Quality of raw materials

Material	Chemical Analysis	Specification	Size	Other properties
Iron Ore(Lumps)	Fe	61.0% min.	10 – 40 mm	Softening Melting range:
	SiO ₂	2.5 ± 0.5 %		1100 - 1400°C
	P	0.10% max.		
	Al ₂ O ₃ /SiO ₂	0.70 max.		
Sinter	Fe	50-58%	5 – 40 mm	RDI(Reduction Degradation Index) <30
	FeO	7-10%		RI(Reducibility Index) >65
	SiO ₂	4-6%		Tumbler Index >70
	Al ₂ O ₃	2-3%		Softening Melting range:
	CaO	9 – 13%		1200 – 1450°C
	MgO	2 – 3%		
Coke	Ash	13 – 15%	25 – 80 mm	CRI(Coke Reactivity Index): 21 -23
	VM(VOLATILE MATTER)	< 1 %		CSR(Coke Strength after Reduction) > 64

	Moisture	5 ± 0.5%		M40 >80%
	S	0.5 - 0.6%		M10 <6%
	FIXED C	82- 85%		
Limestone	CaO	38 % min.	10 – 40 mm	
	SiO ₂	6.5 ± 1%		
	MgO	8.5 ± 0.5%		
LD slag	CaO	40.8 ± 1%	10 – 40 mm	
	MgO	10.5 ± 0.5%		
	SiO ₂	15.50%		
Mn ore	Mn	30% min.	25 – 50 mm	
	SiO ₂	30% max.		
	Al ₂ O ₃	5% max.		
	P	0.30% max.		
CDI coal	FIXED C	60-70%	80 % <90 microns	
	VM(VOLATILE MATTER)	20-25%		
	Ash	9 – 11%		
Quartzite	SiO ₂	96% min	25-50 mm	
	Al ₂ O ₃	1.5% max		

Charging:

High lines and Stock House

High lines: The main responsibility of high lines section is to receive the raw materials required for the production of hot metal from various sources, storing and transporting them to the top of the furnace in time, for the smooth running of the furnace.

Raw materials arriving to the blast furnace department from various sources are unloaded in the RMHP (Raw Material Handling Plant). The ore yard is meant for stocking and averaging of materials. The materials from RMHP are transported to Blast Furnace with the help of wagon tippler, conveyors, stackers and reclaimers.

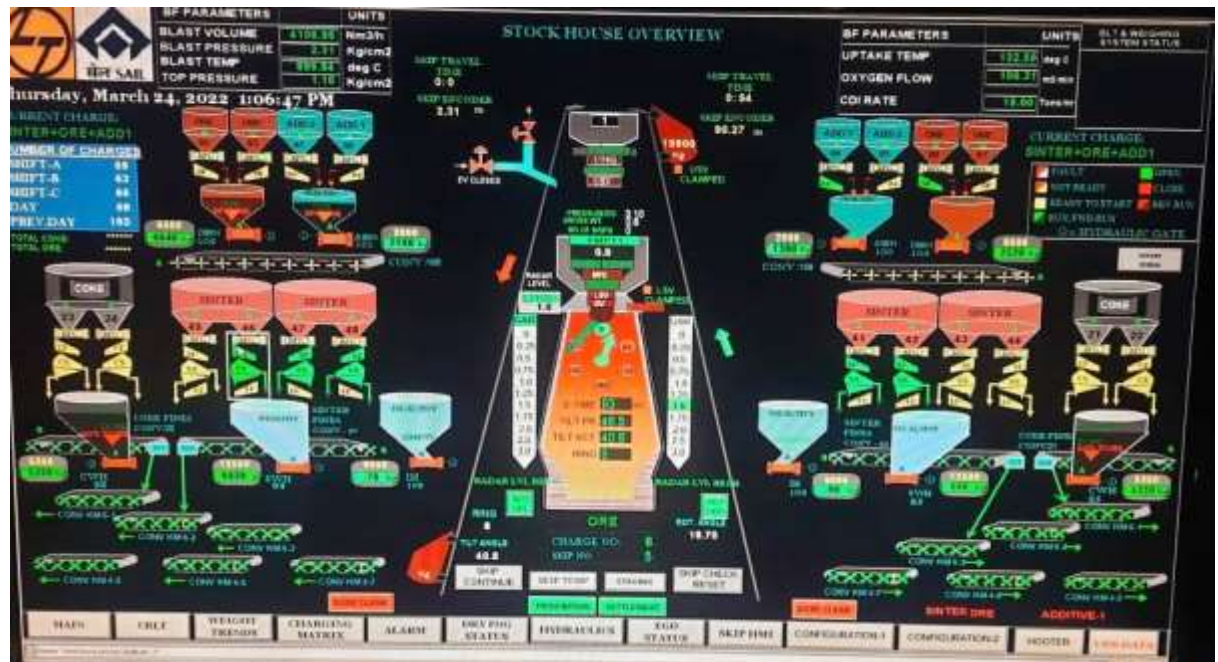
Raw materials from the ore yard are charged by suitable means into the respective bunkers. Alternately in some plants iron ore is received in a wagon Tripler, stack in to piles, and reclaimed using reclaimers.

Sinter from bunker located on the extension tracks of high line is collected in transfer cars moving on rail tracks or sinter comes by means of conveyor belt and is stored in a receiving hopper.

Sinter is screened in stock house, and the fines are returned through conveyor belts.

Coke (25 – 80 mm) from coke sorting plant (CSP) is supplied to the coke bunkers of the blast furnace with the help of conveyor belts and the undersize are returned through conveyor belts.

Stock house: The bunkers are provided with a vibrofeeder, which feeds the material to the conveyor belt/screen. The BF size material is fed to a weighing hopper through ore discharge conveyor. The weighing hopper discharges the material into the skip. There are conveyors to remove the return fines from the system.



Hoist house:

For taking charged materials to the furnace top, two-way skip hoist with 2 skips are provided. The hoist house operates the skip that is driven by two motors. Bell hoist, equalizing valves, test rods etc. are also operated from hoist house.

Flow of material to charging skip are

Bunkers → vibro feeder → conveyor belts → weighing hopper → skip car.

Bunkers → vibro feeder → weighing hopper → skip car.

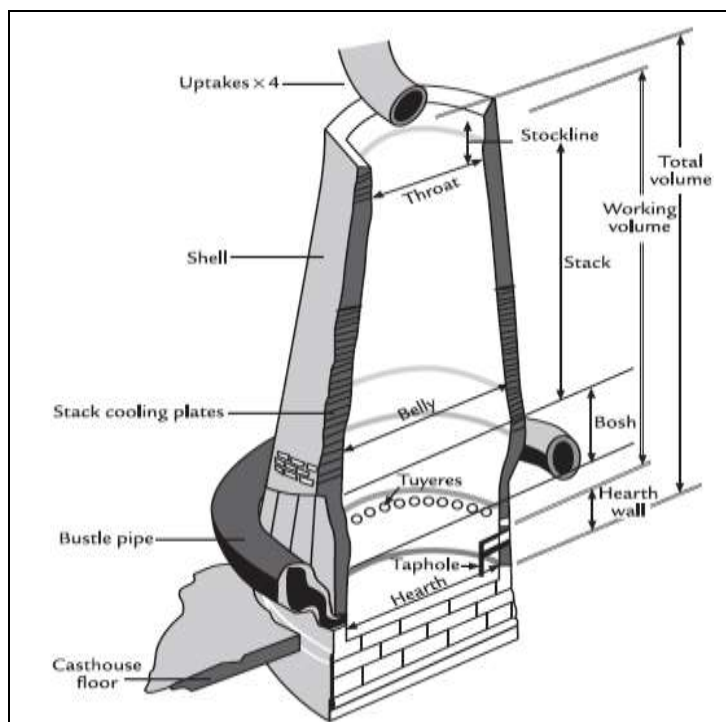
Raw materials including coke are transported and collected into high line bunkers/Stock house placed near the furnaces and then properly screened and weighed. Weighing is done either by scale car or by load cell. These batched proportions of the raw materials are conveyed to the top of the blast furnace via skip car or conveyors and are charged in the blast furnace. The distribution is maintained in such a fashion that alternate layers of coke and iron-containing burden (sinter and iron ore and fluxes) are formed inside the blast furnace.

4.3 Blast Furnace and accessories

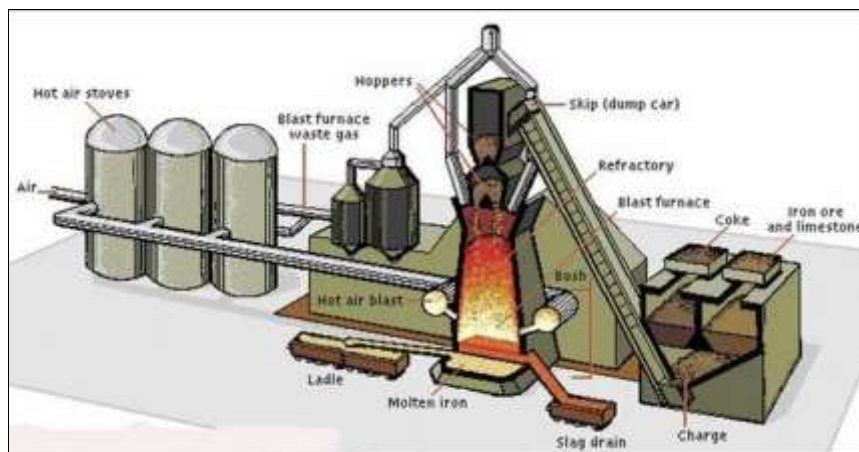
Blast furnace is basically a counter current apparatus, composed of two truncated cones placed base to base.

The sections from top down are:

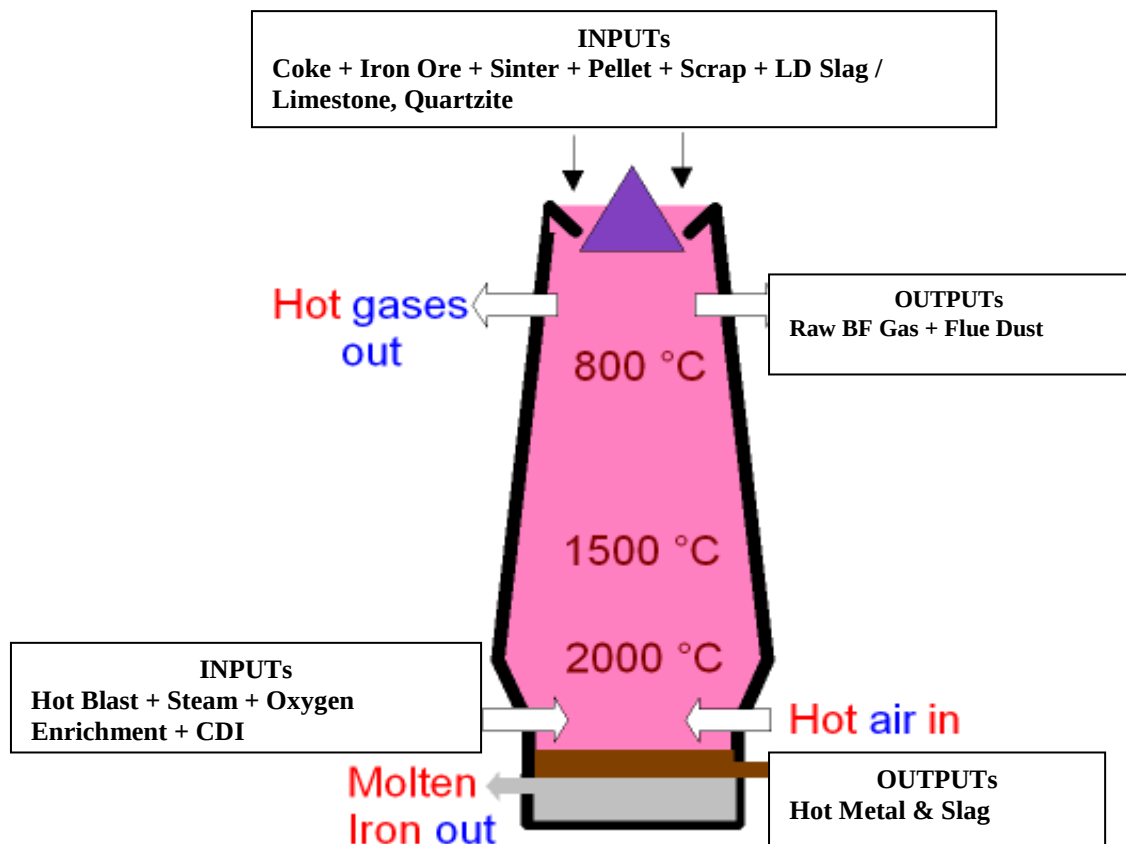
- Throat, where the top burden surface is.
- The shaft or stack, where the ores are heated and reduction starts.
- The bosh parallel or belly, where the softening melting takes place.
- The bosh, where the reduction is completed and the ores are melted down.
- The hearth, where the hot metal and slag is collected and is cast via the tap holes.



BF complex in a nutshell



BF Proper



The entire furnace is lined with suitable refractory and in addition to refractory lining, there are water coolers, designed to enhance the life of the furnaces. **In a blast furnace, fuel, Iron ore, sinter and flux (limestone) are continuously supplied through the top of the furnace**, either through 'double bell system' or 'bell less system'. In the hearth, there is a tap hole of suitable dimension and length, for the purpose of tapping the hot metal.

Since blast furnace is basically a counter current apparatus the descending stream of raw materials extract heat from the ascending stream of gas generated from the burning of coke at the tuyere level. The ascending stream of gas contains CO (carbon monoxide), nitrogen and hydrogen. The ascending reducing gas (CO and H₂) comes in contact with the iron ore thus reduction (this reduction is called indirect reduction) of iron ore takes place at the upper part of the stack (temp less than 900 °C). Coke in the form of C also takes part in the reduction (temp greater than 900 °C) and this reduction is called direct reduction. In the hearth there are multiple tapholes at about 3-4 meter below tuyeres for flushing out hot metal and slag at regular intervals. Tapholes are also extensively water cooled. The number of tap holes, their positioning and dimension will depend upon the capacity of the furnace. Many modern furnaces are having 2 – 4 tap holes.

The furnaces are equipped with tuyeres (water cooled copper construction for admission of hot blast of air) through which preheated air blast at a temperature of about 850 °C – 1200 °C is introduced for burning of coke. Before preheating, the blast of cold air supplied by turbo blowers from power and blowing station and it is introduced into hot blast stoves at a pressure up to 1.8 - 4.5 kg / cm² (gauge pressure) wherein the air is pre – heated. The air blast then passes from the bustle pipe through gooseneck and then tuyere stocks / blow – pipes into tuyeres. The pressure of the blast and its flow rate is dependent upon the capacity of the furnaces and permeability of raw material.

As the stream of the charged material descends down through different temperature zones it gives two products:-

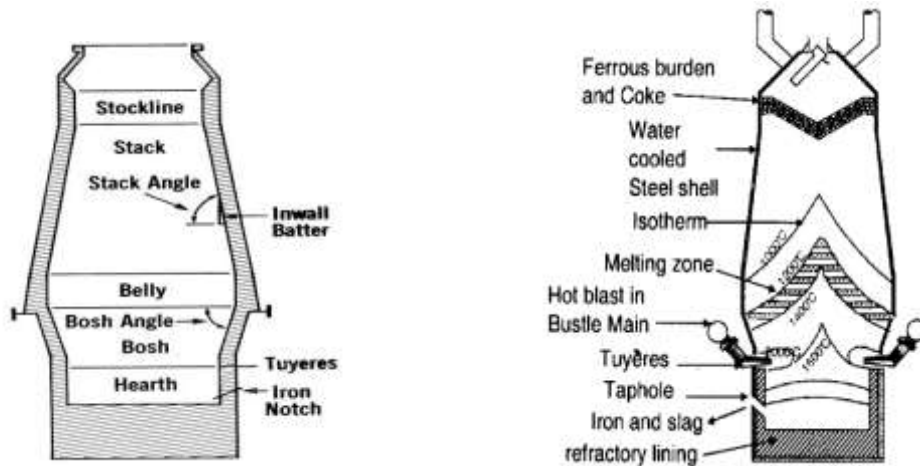
1. Hot Metal in the liquid condition.
2. Slag, in the liquid condition having less density thus floats at the top of metal.

Besides, we get one more important gaseous product from the top of the furnace known as BF gas. It generally comprises of 20 – 24 % CO; 18 - 20 % CO₂, 48-52 % of N₂, H₂ 4-5 %, O₂ 0.1-0.3%. The temperatures of top gases are in the range of 100 – 200 °C.

After cleaning, BF gas is used in blast furnace for stove heating and other area of plant like coke oven heating, and as a mixture with CO gas it is used in refractory materials plant, sintering plant, steel making shop and reheating furnace of rolling mills as a fuel.

Liquid iron collected in the hearth is taken out by opening the tap hole with power driven/Hydraulic drill and oxygen lancing (as per requirement) after regular interval into a train of ladles kept below the runner of the cast house. Slag that comes along with the metal is skimmed off with the help of skimmer plate towards slag runner and collected in slag ladles or to slag granulation plant of cast house (CHSGP). Slag ladles are then sent to the dump yard or slag granulation plant. Metal ladles are either sent to Steel Melting Shop or Pig Casting Machine and Foundry depending upon the requirement.

Schematic Cross Section of the Blast Furnace



Refractory:

Blast furnace is a vertical shaft furnace, enclosed in a welded shell, lined with fire-clay bricks of high alumina content. The hearth bottom, hearth, bosh, belly and the shaft are cooled by means of coolers of various designs. Steel refractory lined plates protect the walls of the furnace top. The bigger furnaces are lined with carbon blocks in the hearth and in the periphery of the hearth bottom. High alumina or Si-Carbide refractory are used in bosh and lower shaft. The design and operation of blast furnace results the high productivity and long life of blast furnaces. The safe and reliable operations are secured by state-of-the-art blast furnace cooling and lining designs

Top charging equipment:

The burden material which reaches to the top of the furnace by skip car or by charging conveyer is to be distributed into the furnace through double bell charging system (Fig-1), rotating charging unit (RCU) (Fig-3) or with Paul-wurth bell less top (BLT) (Fig-2,4) charging system. In BLT charging bells are replaced with charging bins, upper material gate, upper sealing valve, lower material gate and lower sealing valve. This system also has a gearbox to operate the rotating chute. The latter distributes the material inside the furnace periphery in different rings or sector charging, point charging etc. This facilitates better burden distribution inside the furnace as per the “Charging Cyclogram or Pattern” desired by the furnace operator for continuous efficient operation of the furnace.

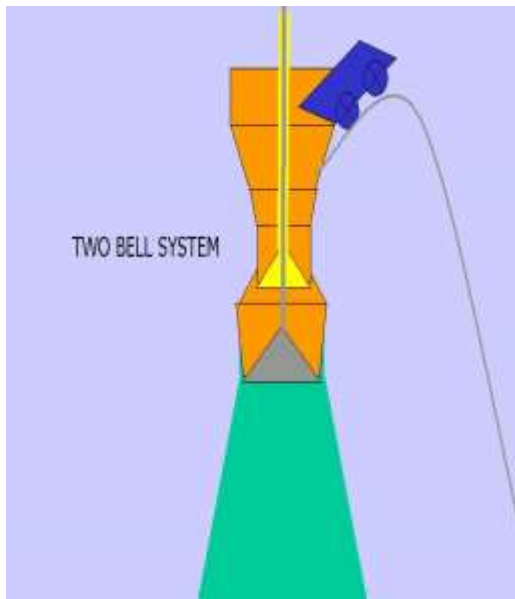


Fig-1. Double Bell

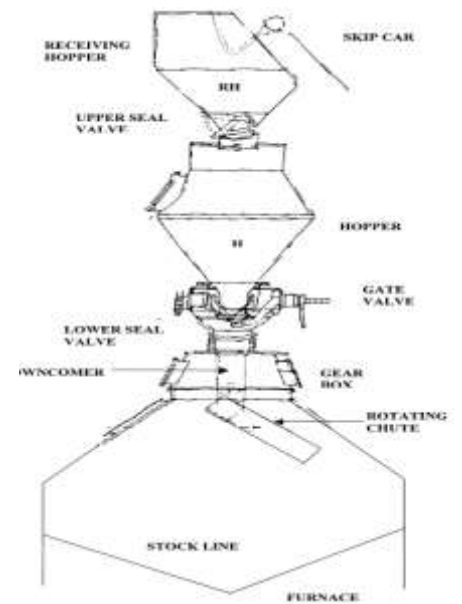


Fig-2 BLT systems



Fig-3 Rotary Charging Unit (RCU)

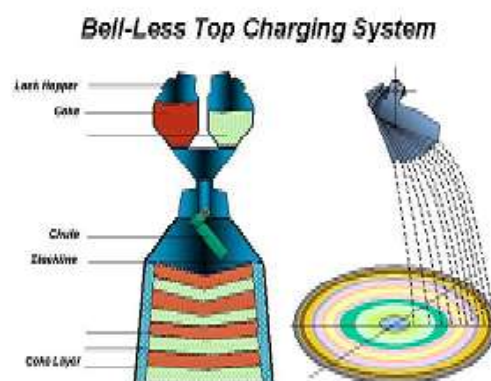


Fig-4 BLT Charging

Charging Sequence: to facilitate smooth working of furnaces, the coke and the non-coke material is to be distributed in a particular fashion in the whole circumference of the blast furnace in accordance to the Charging Cyclogram/ Pattern as determined by the furnace operator. For those different charging sequences is followed. A typical charging sequence is given below:

Sequence 1: COC / COOCC / CCOCC

Sequence 2: CCOO

Each charging cycle consists of 5/6 sequences of either 1 or 2 exclusively or in combination depending on the periphery conditions. Generally in bell-less top furnaces the 2nd sequence is followed i.e. CCOO. C=Coke; O=Non-coke(Ferrous burden) i.e. ore, sinter, Mn ore, Lime Stone or Quartzite etc. The material is distributed in the bf in different rings/sectors as per requirement

Furnace Foreman Control Room (FFCR):

All the activities burden distribution; stoves, cast house, auxiliary fuel injection etc. are controlled from FFCR located in the furnace. Level -0, Level -1 and Level-2(Modern furnaces) automation facilities are there in all BF. All the details regarding the furnace are monitored using HMI(Human machine Interface)/ SCADA (Supervisory Control and Data Acquisition) and mimic panels kept in the FFCR.

Auxiliary Sections:

The auxiliary section of blast furnace consists of following sections:

1. Ladle Repair Shop(LRS)
2. Pig Casting Machine(PCM)
3. Cold Pig Yard (CPY)
4. Clay Mass Shop(CMS)
5. Coal Dust injection Facility(CDI)
6. Cast House Slag Granulation Plant(CHSGP)
7. Slag Dump Yard(SDY)
8. Area Repair Shop (Mech/Elec)
9. Torpedo Ladle Repair Shop(TLRS)

Ladle Repair Shop: Ladle repair shops provided for relining, repairing and cleaning of the iron ladles. Shop contains an EOT cranes for speeding up the job.

Pig Casting Machine: These are double strand pig casting machines. Each machine contains no. of moulds in one belt with lime coating arrangement underneath the machine. Moulds are filled with the hot metal from the ladle at the spout, cooled by water sprays on

the bed while on movement and the pigs are separated from mould chain by knockout arrangement.

EQUIPMENTS of PCM

- Winch to lift the loaded liquid metal ladle.
- Two stands (frames) to hold the ladle firmly, with the paws attached in both the sides of the ladle.
- Runner to receive the liquid metal and to pour into the moulds through spouts.
- Lime spray units to make a thick coating of lime on the moulds to avoid sticking of cold metal with the moulds.
- Individually operated steel belt conveyors to receive the liquid metal and to dispose after pigs are made.
- Chutes to receive the cold pigs and to drop the same into Wagons/Flat car.
- Capstan system at the discharge end to move the loading wagons during pigging.

Cold Pig Yard: Cold pigs from PCM come here. These are stacked according to their quality, and loaded in box wagons with the help of EOT cranes for dispatch to stack yards of customers.

Clay Mass Shop: Here, refractory mass required for blast furnace department is made and stored e.g. mud gun clay, tap hole frame mass and runner mass etc.

Slag Dump Yard: The slag ladles from BF is sent to the dump post for emptying the ladles. Provision exists at the yard for tilting and hammering out the slag with the help of cranes.

CHSGP: Slag granulation plants are attached with the cast houses and the slag generated is granulated at CHSGP. This granulated slag is transported via conveyor belts to the granulated slag yard from where it is sold to the customers (i.e Cement Industries).

Area Repair Shop: Both Mechanical and Electrical Section have their repair shop where necessary supporting repair works are done.

Torpedo Ladle Repair Shop: Torpedo Ladle repair shops provided for relining, repairing and cleaning of the torpedo ladles. Shop contains an EOT cranes, tilting drives for the job.

Auxiliary Fuel Injection

In the present competitive environment, there is a lot of pressure on BF operators to lower the operating costs and maximize **productivity**. One way to achieve this is by injecting auxiliary fuel into the blast furnace. The fuels used for this purpose maybe coal dust, coal tar, natural gas, coal bed methane etc. In SAIL generally coal dust injection (CDI) is being used as auxiliary fuel injection.

The challenge now is to achieve high CDI rates with available quality raw materials, without losing hot metal quality, productivity or BF availability.

Economic and operational benefits achieved by using coal dust injection (CDI) include:

- Lower consumption of expensive coking coals. replacing coke with cheaper soft coking or thermal coals reduces reductant costs;
- Extended coke oven life, since less coke is required to be produced. This is important as many coke ovens are reaching the end of their useful life and significant investment is required to replace or maintain them;
- Higher BF productivity(Ton/m³/Day), that is, the amount of hot metal produced per day (in conjunction with other operational changes);
- Greater flexibility in BF operation. for instance, CDI allows the flame temperature to be adjusted, and the thermal condition in the furnace can be changed much faster than would be possible by adjusting the burden charge at the top of the furnace;
- Improved consistency in the quality of the hot metal and its silicon content;
- Reduced overall emissions, in particular, lower emissions from coke making due to decreased coke requirements.

Gas Cleaning Plant:

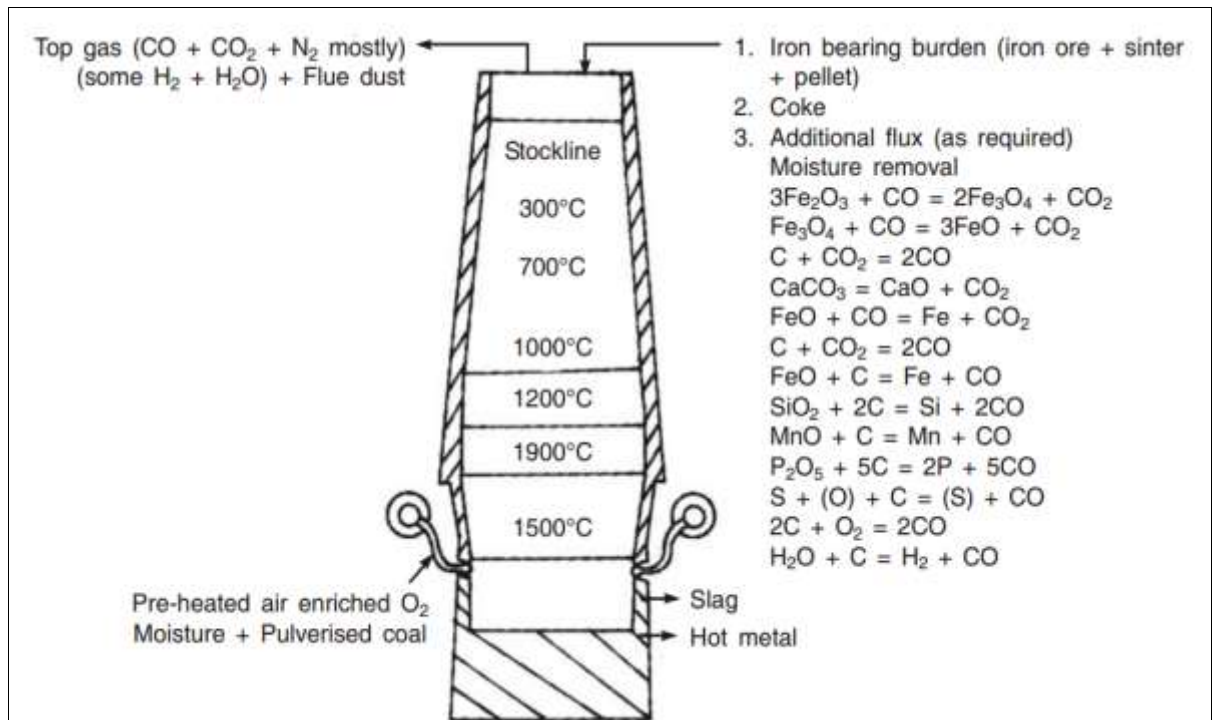
The other product BF gas contains lot of dust in it and it is cleaned in dust catcher, ventury washer and scrubber and finally in electro static precipitator. This activity is done under the supervision of energy management department. The cleaned BF gas is sent to the gas network and is used as a fuel all over the plant.

Flow of BF gas to GCP is

BF Gas → Uptake→Down Comer→Dust Catcher→Ventury Washer→ Scrubber →Electro Static Precipitator→ Cleaned BF Gas →Gas Main

4.4 BF Zones and chemical reactions

Reactions in the Blast Furnace:



UPPER STACK ZONE

- Reduction of Oxides
$$3\text{Fe}_2\text{O}_3 + \text{CO} = 2\text{Fe}_3\text{O}_4 + \text{CO}_2$$
$$\text{Fe}_3\text{O}_4 + \text{CO} = 3\text{FeO} + \text{CO}_2$$
$$\text{FeO} + \text{CO} = \text{Fe} + \text{CO}_2$$
- Carbon Deposition
- Decomposition of Carbonates
- Decomposition of Hydrates Water
- Gas Shift Reaction
$$\text{CO} + \text{H}_2\text{O} = \text{CO}_2 + \text{H}_2$$

MIDDLE STACK ZONE

- Indirect/Direct Reduction
$$\text{FeO} + \text{CO} = \text{Fe} + \text{CO}_2$$
$$\text{CO}_2 + \text{C} = 2\text{CO}$$
$$\text{FeO} + \text{C} = \text{Fe} + \text{CO}$$
- Gas utilization

LOWER STACK ZONE

- Calcinations of Limestone
- Reduction of Various elements
Reduction of unreduced Iron
Reduction of Silicon
- Reduction of Mn, P, Zn etc
- Formation / melting of slag, final reduction of FeO and melting of Fe.

COMBUSTION ZONE

- Burning and combustion of Coke
$$\text{C} + \text{O}_2 = \text{CO}_2 + 94450 \text{ cal}$$
$$\text{CO}_2 + \text{C} = 2\text{CO} - 41000 \text{ cal (solution loss reaction)}$$
- Complete reduction of Iron Oxide

RACEWAY

- Combustion of Coke and Hydrocarbons.
- Combustion of CDI.
- Large evolution of heat.

HEARTH

- Saturation of Carbon with Iron
- Final Reduction of P, Mn, Si and Sulphur

- Reaction impurities reach their final concentrations
- Falling / drop of Metal and Slag bring heat down into the Hearth.

The liquid products hot metal and slag settle in the hearth. These two products are removed periodically from the blast furnace. The process is called tapping the blast furnace.

The golden rule of blast furnace operation is that the furnace conditions should not be disturbed. If for one reason or the other, the quality of charging materials fluctuates, the furnace will be affected. The moisture of coke should be continuously measured and corrective action to be taken. Once the tapping is opened and liquid level begins to fall, the blast pressures drops correspondingly. During the tapping itself, burden descent is fast and irregular. The rise and fall of blast pressure will cause raceway distortions. Similarly, the bosh gas distribution is affected when the burden descent rate increased or decreased. As stock line is not maintained many a time unprepared burden enters the melting zone and increases the thermal requirements. The effect of all these is the disruption of the configuration of the cohesive zone, increase in coke rate and decrease in productivity. Continuous monitoring of the top gas analysis will give an indication about the furnace efficiency.

Common difficulties in operation:

Furnace performance is linked with the smooth operation of the furnace which gets disturbed very often due to various kinds of fluctuations taking place in operating parameters.

Results the number of Irregularities may observed during the operation of furnace like:

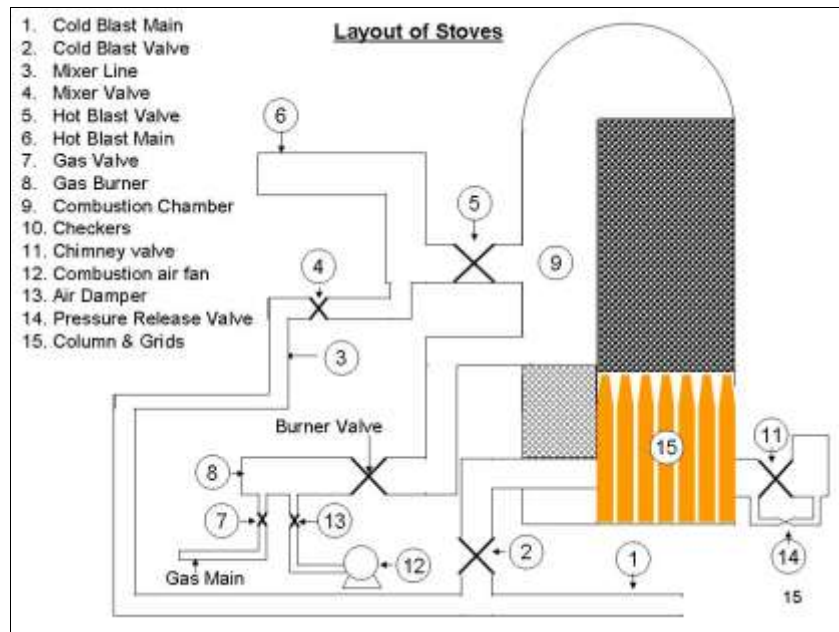
- Channelling
- Scaffolding
- Hanging
- Slipping
- Choking of Hearth
- Chilling of Hearth
- Burning of Tuyeres
- Coke rush through tap-hole.

4.5 Hot blast stoves

The function of Hot blast stove is to preheat the air before admission into the furnace through tuyere. Air is preheated to temperatures between 1000 and 1200 °C in the hot blast stoves.

There are 3 or 4 stoves for each furnace. Each stove consists of a combustion chamber and refractory checker brickwork. Combustion chamber lined with fire bricks and checkers are by alumina brickwork.

For controlling cold and hot blast there are several valves given on the stove. They are: cold blast valve (1), hot blast valve (5), chimney valves (11), by-pass chimney valves (14), gas control valve (7), gas burner (8), and air fan (12).



Hot Blast Stove and its Valve Arrangement

There are two cycles in the stove operation.

1. On gas: stove in the heating mode
2. On blast: stove in the blast mode

In the first cycle the stoves are getting heated by using BF gas and / or coke oven gas. The flue gases (200-350°C) will be carried out through the chimney. This stage is called '**on-gas**'. When the dome temperature reaches to the desired level (1100-1350°C) the gas is stopped and cold blast that is coming from the power and blowing station is sent thorough the cold blast valve, this cycle is called '**on blast**'. The sensible heat that is stored the checker brickwork is carried away by the cold blast and is getting heated. Thus hot blast is produced and this blast is sent into the blast furnace through hot blast valve via hot blast main, bustle pipe, compensator, tuyeres stock and to the tuyeres. The stove kept 'On blast' will continue for $\frac{3}{4}$ hrs - 1 $\frac{1}{4}$ hrs and will be followed by "on gas" cycle (1 $\frac{1}{2}$ hrs – 2 $\frac{1}{2}$ hrs). Thus at any point of time one or two stoves are kept 'on blast' and two stoves are 'on gas' and the cycle is repeated continuously. Heated stove kept isolated, as ready for on blast cycle.

A **snort valve** is located on the cold blast main, regulates the volume of blast. The steam is injected for the humidification of the blast before pre-heating in the stove. Oxygen enrichment is also done whenever necessary through the blast itself. A mixer valve which regulates the flow of cold blast enables to maintain the desired hot blast temperature.

The hot blast reacts with coke and injectants', forming a cavity, called raceway in front of the tuyeres and different reactions takes place in various zone to produce hot metal.

4.6 The Cast House and Slag Granulation Plant

Function

The cast house is the most labor intensive area in the entire blast furnace operation. Its design must be fully integrated with the expected hot metal production, hearth volume, and tapping practice whilst minimizing use of labor, maintenance, materials and improving working environment.

The function of cast house is to tap the liquid metal and slag via the tap hole from hearth on scheduled time and separate the metal and slag by skimmer block with siphon hole in trough which is made up of refractory mass (Castable) and direct metal to metal ladles and slag to the slag ladles or CHSGP.

Process and parts of cast house:

In the BF of single tap hole, there is a provision to flush the slag through the slag notch (called **monkey**) situated at a height of 1400 mm - 1600 mm from the axis of the tap hole. The monkey is equipped with pneumatic or manual cinder stopper. Increasing the number of tapings can reduce flushing operation.

Cast house consist of tap hole, trough, iron and slag runner, rocking runner and their spouts and various equipments (such as EOT crane, Pusher Car, Rocking Runner Tilting Mechanism, Drilling Machine, Mud-gun, JCB / Poklain (excavator) etc). The hot metal is tapped out at an interval of 1-2 hrs depending upon the furnace condition. The tapping time will be around 90 – 120 minutes. Generally 8 -9 tapings will be done in a day. The usual way of opening the tap hole is to drill the tap hole until the skull is reached then oxygen lancing is carried out to melt the skull to get good flow of hot metal.

Generally the tap hole is located in such a way that after tapping minimum amount of metal should remain in the hearth. So it is almost at the bottom most part of the hearth. After opening the tapping hot metal will comes out first. After some time the liquid level in the hearth decreases and the slag that will be floating on the metal comes out of the tap hole. The skimmer plate separates the slag from the metal and diverts the slag into the slag ladles / SGP through slag runners. The hot metal continues to flow down the bend runner from which it is diverted into individual metal ladles. The control of this operation is accomplished by cutters located in the runners or with the help of rocking runner and pusher car. At the end of the tapping the tap hole is closed with the mud gun, which is electrically or hydraulically operated

The hot metal is collected in a refractory lined vessel called hot metal ladle/torpedo ladle and for safety reasons it is filled up to 85 – 90 %. Using these ladles hot metal is transported from blast furnace to mixers in SMS, PCM and foundry as per requirement.

Similarly slag is collected in slag ladles and is dumped in the dump post or sends to slag granulation plants (SGPs) in which slag is granulated, and this granulated slag is sold to cement manufacturers.

The equipments available at the cast house are:

1. Drill Machine → Hydro-pneumatic or electric drilling machines are used for opening the tapping
2. Mud-Gun → Hydraulic or electric drilling machines are used for closing the tapping with anhydrous or water bonded tap hole mass
3. Cast House Crane → for material handling during cast house preparation
4. Rocking runner → to divert the metal into a different metal ladle (tilting runner)
5. Pusher car → used for local placement of the metal ladle

Analysis of hot metal, slag and top gas

Hot Metal		Slag		BF gas	
Si	0.6 - 0.8 %	SiO ₂	34 – 36 %	CO	20 – 24 %
Mn	0.05 - 0.10 %	Al ₂ O ₃	16 – 20 %	CO ₂	18 – 20 %
S	0.050 % max.	CaO	34 – 36 %	N ₂	48 – 52 %
P	0.05-0.15 %	MgO	8 – 10 %	H ₂	4 – 5 %
C	4 – 5 %	MnO	<1%		
		Basicity: CaO/SiO ₂	0.98-1.00		

Modern technological developments

Some of the modern technological developments implemented at our plants are:

Beneficiation - To upgrade the quality of iron ore, special emphasis is for preferential removal of alumina from the gangue.

Bedding, Blending, Sizing and Screening of burden - Physical and chemical characteristic of iron ore, coal and limestone vary from deposit to deposit and also from one mine to another. For trouble-free operation of blast furnaces, it is essential to ensure supply of raw materials of consistent and uniform quality. The bedding and blending of the incoming raw is adopted before processing them.

Use of 70 – 80 % sinter in the burden - It has been proved that with the use of sinter in the burden the productivity of blast furnace increases. Along with sinter 10-15% pellet in burden mix will also give additional benefits.

Conveyor charging - All burden materials are delivered to the furnace top by conveyor. This is economical for bigger blast furnaces. All 3 bigger blast furnaces in SAIL i.e. BF-5ISP (4161m³), BF-5 RSP (4060m³) and BF-8 BSP (4060m³) have conveyor charging facilities.

Bell less top - In place of conventional two bell charging system, two charging hoppers with rotating chute are installed. The rotating chute distributes the material in the desired manner. The system is easy to maintain. The system has been adopted in BF - 4, 5, 6, 7, 8 of BSP, BF # 1, 4 & 5 of RSP, BF # 3 of DSP and all the blast furnaces of BSL.

Movable throat armour - This is installed along with two-bell system. The distribution of material is controlled by positioning the throat armour at proper location. The system improves the burden distribution. BF # 2 & 4 of DSP has been provided with this system.

Amanoscope - The device is fitted at the top of the furnace. It emits a beam of infrared rays over the material surface of the stock and takes the photograph.

Furnace probes - Probes are fitted above (above burden probe) the stock level / below the stock level (under burden probe) in order to monitor temperature distribution and collect samples of burden material and gas.

Cast House Slag Granulation - In this design the liquid slag from cast house runner is led to the granulating unit located very near to the cast house. This would eliminate the need for maintenance of large fleet of slag ladles, reduce the cost of production, avoid delays and increase the yield of granulated slag. BF # 4, 5, 6, 7 & 8 of BSP is having cast house slag granulation. This facility is installed in all furnaces of BSL. This facility already exists in BF # 1, 4 & 5 of RSP and BF # 3 & 4 of DSP.

Slag Granulation Plant marked by the following features:

- It facilitates dry tapping of the furnace; not being limited by ladles availability.
- Utilization of molten slag is very high (98%) compared to distant granulation (70%) and better slag granules quality. Safe, efficient and pollution free working environment by avoiding movement of ladles etc

The main advantages of CHSGP are:

1. Very compact and requires less space.
2. Fully automatic, less manpower requirement.
3. Low electricity & compressed air consumption.
4. Completely covered installation from granulation unit to the dewatering station with connection to stack for the collection of steam & fumes and venting out the same to the atmosphere at high level

Coal Dust Injection - Non-coking coal is injected through tuyere using nitrogen as carrier. This reduces the coke rate and thus saves the valuable coking coal, which is also not abundantly available in India. Coal dust injection is normally associated with high blast temperature and oxygen enrichment. All furnaces of SAIL plants have been provided with a coal dust injection system.

External Desulphurization of Hot Metal - With the introduction of continuous casting technology and increased demand for high quality steel, requirement of low Sulphur (less than 0.025%) hot metal has increased. For this purpose hot metal from BF is desulphurised by injecting desulphurising agents such as calcium carbide, lime soda ash and magnesium in the hot metal ladle. One desulphurising unit has been installed at RSP, ISP, BSP & BSL (under installation).

Cast House Desiliconisation -Silicon from hot metal is partially removed by adding mill-scale, iron ore, along with lime in the hot metal runner. Such installations are working in Abroad.

De-Phosphorisation of Hot Metal – De-phosphorising agents like soda ash and lime based flux are added in hot metal in transport vessel to reduce phosphorous content of hot metal.

Automation & Computer control - In case of fully automatic operation, the computer (HMI system) is connected to PLC (Programmable Logic Controller) which receives signals from various sensors and determines the optimum point values and commands the equipments to operate automatically. Automatic control of charging & stoves are provided in furnaces.

As per the decision of SAIL management, to achieve targeted hot metal production the following measures have been envisaged in the blast furnace area.

1. Modernization/ Up gradation of the BF with respect to refractory, cooling system, stoves ,auxiliary fuel injection and supporting equipments
2. Installation of a new and bigger furnace at a separate location along with a new stock house and new material handling facilities with modified sinter plant and coke ovens. New furnace has been commissioned in RSP, ISP and BSP.
3. Modification of the existing material handling system.
4. Introduction of torpedo ladles.
5. Improvement of logistics in blast furnace area etc.
6. Waste Heat Recovery System (recovery of heat from outgoing stove hot flue gases to heat up the Combustion Air and BF Gas and thus enhance the heating of stoves).
7. TRT (Top Gas Recovery Turbine to generate Power) is provided in new furnaces at ISP,RSP and BSP @ 14 MW power generation per day
8. Radar Stock Level Indicator enables to measure the stock level when the furnace is off-rod (beyond 3 meters) is provided in BF # 2 of BSL, BF # 6 of BSP and new furnaces at ISP,RSP and BSP

Techno-economics:

Productivity: the amount of hot metal produced per cubic meter of the furnace volume in a day, tones /m³/day (either on working volume or useful volume basis)

Fuel rate: The amount of fuel required to produce one tone of hot metal, kg/thm. It includes coke rate + aux. fuel rate + nut coke rate (added in sinter). Carbon rate is more appropriate measure as carbon content of the fuel varies from time to time.

- 1% reduction in coke ash reduces the coke rate by 8 – 10 kg/thm (reduction of coke rate gives an overall saving of 0.5 % of the total energy consumption of steel plant)
- 1% reduction in gangue of iron ore results in coke saving of 1.5 %.
- 100 °C increase in HBT (Hot Blast Temperature) results in coke saving of 2 – 3 %.

Energy consumption for an iron making:-

Energy Consumption	Area Wise Energy Consumption
Coke Making	18 – 19 %
Iron Making & Sinter Making	49 – 50 %
Steel Making	4. – 9 %

4.7 Safety and environment

All applicable safety standards must be followed.

The use of PPEs (Personal Protective Equipment) like safety helmet, safety shoes, hand gloves, gas masks, heat resistant jackets/coats, goggles and dust masks are to be used religiously while working in different areas of Blast furnace

Laid down procedures like “permit to work” are to be strictly followed before taking any shut-down of equipment for maintenance. The stipulated HIRA (hazard Identification & Risk Assessment), WI (Work Instruction), SOPs (Standard Operating Practices) and SMPs (Standard Maintenance Practices) should be adhered strictly.

Persons should be cautious about the gas prone areas and should know about the gas hazards. EMD clearance is a must before taking up any job in gas lines or gas prone areas.

Following do's and don'ts are to be followed for safe Operation of blast furnace.

DO's

1. Ring bell/hooter during crane movement
2. Safe distance should be maintained while looking through Tuyeres (wear safety glass)
3. Before putting any stove to gas mode if any gas leakage observed then remove the agency working in stove platform
4. Always carry CO-GAS monitor & gas safety man while going top of the Furnace for checking any abnormalities & during stove area inspection
5. Always monitor proper gas burning in tapholes, monkey, tuyeres, tuyere Coolers during running of Furnace

6. Shut Down Work for repair etc should be carried as per procedure (protocols) duly approved.
7. Permit to Work (PTW) must be taken before any work on equipment
8. All applicable safety standards must be followed.

Don'ts

1. Don't allow any unauthorized person on stove platform.
2. Don't allow any one in Cast House area of blast furnace without safety appliances.
3. During any gas leakages don't allow anyone in the Cast House area & stove platform.
4. Don't Operate lift without proper knowledge of Mech./Elect. Operation of Lift, Always ask for lift--operator.
5. MCC panels should be operated only by authorized personnel.
6. Don't allow any work without proper Permit to Work (PTW).
